Quantum Communication Lab:
Sending Messages with Quantum Mechanics

Student Lab Handout

Duration: 1h15
Format: Pair work, guided coding lab

Learning Objectives

By the end of this lab, you will be able to:

- Explain how entanglement acts as a resource for communication
- Implement superdense coding to transmit classical information
- Describe quantum teleportation as state transfer, not particle transfer
- Distinguish classical and quantum communication constraints
- Interpret the role of measurement and classical bits in quantum protocols

Session Overview

In this lab, you will explore two canonical protocols from *quantum information theory*:

- **Superdense Coding**: sending two classical bits using a single qubit
- **Quantum Teleportation**: transferring an unknown quantum state using entanglement and classical communication

Rather than focusing on speedups or algorithms, this lab highlights how quantum mechanics enables *new modes of information transfer*.

Part A: Superdense Coding

Concept

Alice and Bob share an entangled Bell pair prepared in advance. Alice wants to send **two classical bits** to Bob but is allowed to transmit **only one qubit**.

Classically, this is impossible. Quantum mechanically, it is achievable using entanglement.

Bell Pair Preparation

The shared resource is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

created using a Hadamard gate followed by a CNOT.

```
def create_bell_pair():
    qc = QuantumCircuit(2)
    qc.h(0)
    qc.cx(0, 1)
    return qc
```

Alice's Encoding

Alice encodes a two-bit message by applying a local operation to her qubit:

Message	Gate(s) applied by Alice
00	Identity
01	X
10	Z
11	XZ

Mathematical Description of Alice's Encoding Alice and Bob initially share the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big),$$

where the first qubit belongs to Alice and the second qubit belongs to Bob.

Alice encodes her two-bit classical message by applying a single-qubit Pauli operator $U \in \{I, X, Z, XZ\}$ to *her* qubit only. Mathematically, this corresponds to the action of the operator $(U \otimes I)$ on the joint state.

We now derive explicitly how each operation transforms the shared state.

Message `00`: Identity Alice applies the identity operator I:

$$(I \otimes I)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big) = |\Phi^+\rangle\,.$$

Message `01`: Bit Flip (X) Recall that the Pauli-X operator acts as

$$X\,|0\rangle = |1\rangle\,, \qquad X\,|1\rangle = |0\rangle\,.$$

Applying X to Alice's qubit gives

$$\begin{aligned}(X \otimes I)\,|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\big(X\,|0\rangle\,|0\rangle + X\,|1\rangle\,|1\rangle\big) \\ &= \frac{1}{\sqrt{2}}\big(|10\rangle + |01\rangle\big) = |\Psi^+\rangle\,.\end{aligned}$$

Message 10: Phase Flip (Z) The Pauli-Z operator acts as

$$Z|0\rangle = |0\rangle, \qquad Z|1\rangle = -|1\rangle.$$

Applying Z to Alice's qubit yields

$$\begin{aligned}(Z \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\big(Z|0\rangle|0\rangle + Z|1\rangle|1\rangle\big) \\ &= \frac{1}{\sqrt{2}}\big(|00\rangle - |11\rangle\big) = |\Phi^-\rangle.\end{aligned}$$

Message 11: Bit and Phase Flip (XZ) Finally, Alice applies XZ. Using

$$XZ|0\rangle = |1\rangle, \qquad XZ|1\rangle = -|0\rangle,$$

we find

$$\begin{aligned}(XZ \otimes I)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}\big(XZ|0\rangle|0\rangle + XZ|1\rangle|1\rangle\big) \\ &= \frac{1}{\sqrt{2}}\big(|10\rangle - |01\rangle\big) = |\Psi^-\rangle.\end{aligned}$$

By applying only a single-qubit operation locally, Alice maps the shared entangled state $|\Phi^+\rangle$ onto one of the four mutually orthogonal Bell states:

Classical message	Resulting Bell state
00	$\|\Phi^+\rangle$
01	$\|\Psi^+\rangle$
10	$\|\Phi^-\rangle$
11	$\|\Psi^-\rangle$

Because these states form an orthonormal basis, Bob can later distinguish them perfectly using a joint measurement, allowing him to recover Alice's two classical bits with certainty.

```python
def encode_message(qc, message):
    if message == "01":
        qc.x(0)
    elif message == "10":
        qc.z(0)
    elif message == "11":
        qc.z(0)
        qc.x(0)
    return qc
```

Bob's Decoding

After Alice sends her qubit, Bob performs the inverse Bell operation to decode the message.

Derivation of Bob's Decoding After Alice sends her qubit, Bob holds *both* qubits of the entangled system. Depending on Alice's encoding, the joint state is one of the four Bell states:

$$|\Phi^+\rangle, \quad |\Psi^+\rangle, \quad |\Phi^-\rangle, \quad |\Psi^-\rangle.$$

Bob's task is to determine *which* Bell state he has received, in order to recover Alice's original two classical bits.

To do so, Bob applies a fixed decoding circuit consisting of:

- a CNOT gate with qubit 0 as control and qubit 1 as target;
- followed by a Hadamard gate on qubit 0.

We now show explicitly how this decoding circuit maps each Bell state to a distinct computational basis state.

Decoding $|\Phi^+\rangle$ Start with

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Applying the CNOT:

$$\text{CNOT}\,|00\rangle = |00\rangle\,, \qquad \text{CNOT}\,|11\rangle = |10\rangle\,,$$

so the state becomes

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Next, applying the Hadamard gate to the first qubit,

$$H\,|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H\,|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

we obtain

$$|00\rangle\,.$$

Decoding $|\Psi^+\rangle$ Start with

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle).$$

Applying the CNOT:

$$\text{CNOT}\,|10\rangle = |11\rangle\,, \qquad \text{CNOT}\,|01\rangle = |01\rangle\,,$$

giving

$$\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle).$$

Applying the Hadamard to the first qubit yields

$$|01\rangle\,.$$

Decoding $|\Phi^-\rangle$ Start with

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle).$$

Applying the CNOT:

$$\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle).$$

Applying the Hadamard transform gives

$$|10\rangle\,.$$

Decoding $|\Psi^-\rangle$ Start with

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle).$$

After the CNOT:

$$\frac{1}{\sqrt{2}}(|11\rangle - |01\rangle).$$

Applying the Hadamard yields

$$|11\rangle\,.$$

Bob's fixed decoding circuit maps the Bell basis onto the computational basis:

Bell state received	Measurement outcome
$\|\Phi^+\rangle$	$\|00\rangle$
$\|\Psi^+\rangle$	$\|01\rangle$
$\|\Phi^-\rangle$	$\|10\rangle$
$\|\Psi^-\rangle$	$\|11\rangle$

Thus, by performing this decoding operation followed by a measurement in the computational basis, Bob deterministically recovers the two classical bits sent by Alice.

```python
def decode_message(qc):
    qc.cx(0, 1)
    qc.h(0)
    return qc
```

Execution

```python
message = "11"

circuit = create_bell_pair()
encode_message(circuit, message)
decode_message(circuit)
circuit.measure_all()
```

Note: Qiskit reports measurement outcomes in little-endian order. To compare with the original message, reverse the printed bit string.

Extension: Sending a Text Message

Using superdense coding repeatedly, Alice can transmit a full text message. Each character is first encoded in 8-bit ASCII, grouped into 2-bit blocks, and sent sequentially.

> **Remark:** Superdense coding reduces the number of transmitted *qubits*, not the amount of transmitted *classical information*.

Note We use 8-bit blocks because they are the standard size for a byte, which is the fundamental unit of data in modern computing. The ASCII Standard: The most common character encoding (ASCII) originally used 7 bits to represent 128 characters (A-Z, 0-9, etc.). Computers quickly moved to 8 bits to add an extra bit for "extended ASCII" or error checking, making 1 byte per character the universal rule for basic text. In the modern world, 8 bits isn't always enough. While ord('A') is 65 (1 byte), an emoji 😂 has a much higher Unicode value (128514) that requires 21 bits. When you use your code on an emoji, format(..., '08b') will actually break its 8-bit constraint and return a longer string (like 1111101011000010), which can mess up your formatting if you're expecting fixed-width blocks.

Exercises: Sending and Receiving Messages

In this exercise, you will use superdense coding to *send* and *decode* classical messages using quantum circuits.

Work in pairs. For each task, verify that the decoded message matches the expected output exactly.

Exercise 1: Send a Classical Message

Alice would like to send the message

``hello''

to Bob using superdense coding.

(a) Convert the string ``hello'' into its binary representation using 8-bit ASCII encoding.

(b) Group the resulting binary string into *pairs of bits*. Each 2-bit block will correspond to a single execution of the superdense coding protocol.

(c) For each 2-bit block:

- Alice encodes the block by applying the appropriate Pauli operation to her qubit.
- Alice sends her qubit to Bob.
- Bob applies the decoding circuit and measures both qubits.

(d) Reconstruct the original binary string from Bob's measurement outcomes, and convert it back into text.

(e) Verify that Bob recovers the message ``hello'' exactly.

Check: How many qubits are physically transmitted from Alice to Bob? How many classical bits does the message contain?

Exercise 2: Decode a Given Binary Message

Bob receives the following binary string, which was transmitted using superdense coding:

01001000 01100001 01110000 01110000 01111001 00100000
01101000 01101111 01101100 01101001 01100100 01100001
01111001 01110011 00100001

(a) Divide the binary string into 2-bit blocks.

(b) Use the superdense coding decoding procedure to recover each 2-bit block.

(c) Concatenate the decoded blocks into a single binary string.

(d) Group the result into 8-bit bytes and convert each byte into its ASCII character.

(e) Write down the decoded message in plain text.

Check: Does the decoded message form a meaningful English sentence?

Reflection

Answer briefly:

1. Why is entanglement necessary for superdense coding?
2. What limits the speed of information transfer in this protocol?
3. In what sense does superdense coding reduce the number of transmitted *qubits*, but not the amount of transmitted *information*?

Part B: Quantum Teleportation (Optional)

Concept

Quantum teleportation transfers the *state* of a qubit, not the physical qubit itself, using:

- One shared entangled pair
- Two classical bits
- Local operations and measurements

No information travels faster than light.

Teleportation Circuit

Teleportation requires three qubits:

- Qubit 0: the state to be teleported (Alice)
- Qubit 1: Alice's half of the Bell pair
- Qubit 2: Bob's half of the Bell pair

```
def quantum_teleportation_circuit(initial_state_gate=None):
    # 3 qubits: q0 (Alice's state), q1 (Alice's ebit), q2 (Bob's ebit)
    # 3 classical bits to store all measurements
    qc = QuantumCircuit(3, 3)

    # STEP 0: Prepare the state to be teleported on q0
    # Example: applying an X gate transforms |0> to |1>
    if initial_state_gate:
        initial_state_gate(qc, 0)
    qc.barrier()

    # STEP 1: Create entanglement between Alice (q1) and Bob (q2)
    qc.h(1)
    qc.cx(1, 2)
    qc.barrier()

    # STEP 2: Alice's operations (Bell-state measurement)
    qc.cx(0, 1)
    qc.h(0)
    qc.barrier()

    # STEP 3: Alice measures her two qubits
    qc.measure(0, 0)
    qc.measure(1, 1)
    qc.barrier()
```

```
26
27      # STEP 4: Bob's conditional corrections
28      # If q1 measurement was 1, Bob applies X. If q0 was 1, Bob applies Z.
29      with qc.if_test((qc.clbits[1], 1)):
30          qc.x(2)
31      with qc.if_test((qc.clbits[0], 1)):
32          qc.z(2)
33
34      return qc
35
36  # --- Execution and Verification ---
37  # We teleport the |1> state by applying an X gate at the start
38  def prepare_state(circ, target): circ.x(target)
39
40  teleport_qc = quantum_teleportation_circuit(prepare_state)
41  teleport_qc.measure(2, 2) # Measure Bob's qubit to verify
42
43  backend = AerSimulator()
44  job = backend.run(transpile(teleport_qc, backend), shots=1024)
45  counts = job.result().get_counts()
46
47  # We check bit 2 (the leftmost bit) to see if Bob received the |1> state
48  print(f"Teleportation␣Results:␣{counts}")
49  plot_histogram(counts)
```

Mathematical Derivation of Quantum Teleportation

Quantum teleportation transfers the *state* of a qubit from Alice to Bob without physically sending the qubit itself. The protocol relies on entanglement, measurement, and classical communication.

We label the qubits as follows:

- Qubit 0: the unknown state held by Alice,
- Qubit 1: Alice's half of the entangled pair,
- Qubit 2: Bob's half of the entangled pair.

Step 0: Initial State

Alice begins with an arbitrary single-qubit state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

which she wishes to teleport to Bob.

The three-qubit system initially is

$$|\psi\rangle_0 \otimes |0\rangle_1 \otimes |0\rangle_2\,.$$

Step 1: Shared Entanglement

Alice and Bob create a Bell pair between qubits 1 and 2 by applying a Hadamard gate followed by a CNOT gate:

$$|\Phi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The joint three-qubit state becomes

$$|\Psi_0\rangle = (\alpha\,|0\rangle + \beta\,|1\rangle)_0 \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{12}.$$

Step 2: Alice's Bell-State Measurement

Alice now performs a CNOT gate with qubit 0 as control and qubit 1 as target, followed by a Hadamard gate on qubit 0.

After these operations, the combined state can be rewritten as

$$|\Psi_1\rangle = \frac{1}{2}\Big(|00\rangle_{01} (\alpha |0\rangle + \beta |1\rangle)_2 + |01\rangle_{01} (\alpha |1\rangle + \beta |0\rangle)_2 + |10\rangle_{01} (\alpha |0\rangle - \beta |1\rangle)_2 + |11\rangle_{01} (\alpha |1\rangle - \beta |0\rangle)_2 \Big).$$

At this point, qubits 0 and 1 are maximally entangled with Bob's qubit 2.

Step 3: Alice's Measurement

Alice now measures qubits 0 and 1 in the computational basis.

This produces two classical bits,

$$m_0, m_1 \in \{0, 1\},$$

and collapses Bob's qubit into one of four related states:

Measurement $(m_0 m_1)$	Bob's state	Required correction
00	$\alpha \|0\rangle + \beta \|1\rangle$	I
01	$\alpha \|1\rangle + \beta \|0\rangle$	X
10	$\alpha \|0\rangle - \beta \|1\rangle$	Z
11	$\alpha \|1\rangle - \beta \|0\rangle$	XZ

Step 4: Bob's Conditional Corrections

Alice sends the two classical bits (m_0, m_1) to Bob using a classical channel.

Bob applies the corresponding correction operator:

$$X^{m_1} Z^{m_0},$$

which deterministically transforms his qubit into

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle .$$

Thus, Bob recovers *exactly* the original quantum state held by Alice.

Key Observations

- The original state is destroyed by Alice's measurement (no cloning).
- Two classical bits are required to complete the protocol.
- Entanglement enables state transfer, but cannot transmit information faster than light.

Discussion

1. Why are two classical bits necessary for teleportation?
2. Why can Alice not keep a copy of the state?
3. Compare the resources required for superdense coding and teleportation.
4. Why does teleportation not allow faster-than-light communication?

Exercise 3: Teleporting a Text Message

You are now given a function that teleports a classical message by teleporting *each bit individually.*

(a) Convert the string

``Hello''

into its binary representation using 8-bit ASCII encoding.

(b) For each bit in the binary string, use the quantum teleportation protocol to transmit the corresponding quantum state from Alice to Bob.

(c) Reconstruct the received binary string from Bob's measurement outcomes.

(d) Convert the binary string back into text and verify that Bob recovers the original message exactly.

Check: How many entangled Bell pairs are consumed when teleporting the message ``Hello''? How many classical bits are sent?

Key Takeaway

> Entanglement does not transmit information by itself but when combined with classical communication and measurement, it enables fundamentally new ways of sending information.